

8 - Estimators of the Parameters of a Gaussian Distribution. ML Estimators. The Distribution of the Sample Mean and the Sample Covariance. The Wishart Distribution and its Main Properties. Law of Large Numbers and Central Limit Theorem.

Applied Statistics

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1 Big Picture and Motivation

- We observe data $X_1, X_2, \dots, X_n$ and we believe it comes from a Gaussian distribution with **unknown** parameters μ and Σ .
- **Central question:** How do we estimate μ and Σ from a finite sample? And once we have our estimates, how good are they?
- This question drives the entire lecture. It proceeds in four steps:
 1. **Setup:** Define the model and the intuitive “obvious” estimators.
 2. **MLE:** Derive estimators rigorously from the maximum likelihood principle.
 3. **Exact distributions:** Find what distribution $\bar{X}$ and S follow under Gaussianity (requires the Wishart distribution).
 4. **Asymptotics:** Describe what happens as $n \rightarrow \infty$ — without requiring Gaussianity (LLN and CLT).
- **Key distinction to keep in mind throughout:**
 - **Exact results** (Wishart, χ^2): valid for any finite n , require Gaussian data.
 - **Asymptotic results** (LLN, CLT): valid only as $n \rightarrow \infty$, but require only finite mean and covariance — no Gaussianity needed.
 - *Exam rule:* “Normally distributed data” $\Rightarrow$ exact results. “Large sample” or “general distribution” $\Rightarrow$ CLT/LLN.

2 Setup — The Multivariate Gaussian Model

- Let $\underline{X} \sim N_p(\mu, \Sigma)$, a p -dimensional random vector with:

- Mean vector: $\mu \in \mathbb{R}^p$
- Covariance matrix: $\Sigma \in \mathbb{R}^{p \times p}$, symmetric and positive definite
- Each observation $X_i \in \mathbb{R}^p$ is a column vector:

$$X_i = \begin{bmatrix} x_{i1} \\ \vdots \\ x_{ip} \end{bmatrix}, \quad x_{ij} \in \mathbb{R}$$

- We observe n independent and identically distributed (iid) realisations:

$$X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} N_p(\mu, \Sigma)$$

- **Goal:** Estimate the unknown μ and Σ given only the data $X_1, \dots, X_n$.
- **Obvious estimators** (intuitive, before any formal derivation):
 - μ is estimated by the **sample mean**:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \longrightarrow \quad \text{unbiased: } \mathbb{E}[\bar{X}] = \mu$$

- Σ is estimated by the **sample covariance matrix**:

$$S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^\top \quad \longrightarrow \quad \text{unbiased: } \mathbb{E}[S] = \Sigma$$

- The $n-1$ denominator is **Bessel's correction**: it compensates for the fact that $\bar{X}$ is estimated from the same data, which makes deviations $(X_i - \bar{X})$ systematically smaller than the true deviations $(X_i - \mu)$ would be.
- **Intuition:** Just as the sample mean is the natural estimate of the population mean in 1D, $\bar{X}$ is its multivariate extension. And just as sample variance divides by $n-1$ to be unbiased in 1D, so does S . The multivariate case is a direct analogue.

3 Maximum Likelihood Estimation (MLE)

3.1 Motivation and Definition

- The obvious estimators are intuitively compelling, but we want a **principled, general method** that works for any model.
- **MLE principle:** Choose the parameter value that makes the observed data most probable (or most dense, for continuous distributions).
- **Likelihood function:** For a parameter θ and observed data, the likelihood is:

$$L(\theta \mid \text{data}) = P(\text{data} \mid \theta)$$

Viewed as a function of θ for fixed data (not as a probability over θ).

- **MLE definition:**

$$\hat{\theta} = \arg \max_{\theta} L(\theta \mid \text{data})$$

- In practice, we always maximise the **log-likelihood** $\ell(\theta) = \log L(\theta \mid \text{data})$, because:
 - Products of densities become sums — far easier to differentiate.
 - The exponential in the Gaussian density cancels with the logarithm.
 - The argmax is unchanged: log is strictly monotone increasing, so the maximiser is the same.

3.2 Warm-Up: Bernoulli MLE

- Before tackling the Gaussian, we build intuition with the simpler Bernoulli case.
- Let $X \sim \text{Bernoulli}(p)$, with unknown $p \in [0, 1]$. The distribution is:

$$X = \begin{cases} 0 & \text{with probability } 1 - p \\ 1 & \text{with probability } p \end{cases}$$

- Observe $n = 5$ iid realisations: $X_1 = 0, X_2 = 0, X_3 = 1, X_4 = 0, X_5 = 1$.
- **Likelihood:** Since observations are iid, multiply individual probabilities:

$$L(p \mid \text{data}) = P(X_1 = 0, X_2 = 0, X_3 = 1, X_4 = 0, X_5 = 1) = (1-p)(1-p)p(1-p)p = p^2(1-p)^3$$

- In general, for n iid Bernoulli observations with $\sum_{i=1}^n X_i$ observed successes:

$$L(p \mid X_1, \dots, X_n) = p^{\sum_{i=1}^n X_i} (1-p)^{n-\sum_{i=1}^n X_i}$$

- The likelihood function maps $L : [0, 1] \rightarrow [0, 1]$. We find $\hat{p}$ at its peak.
- **Log-likelihood:**

$$\ell(p) = \left(\sum_{i=1}^n X_i \right) \log p + \left(n - \sum_{i=1}^n X_i \right) \log(1-p)$$

- **MLE:** Set $\frac{d\ell}{dp} = 0$ and solve:

$$\hat{p} = \arg \max_{p \in [0,1]} L(p \mid \text{data}) = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

In our example: $\hat{p} = 2/5 = 0.4$.

- **Intuition:** Before computing any formula, ask yourself — “what value of p would make this data most plausible?” With 2 ones and 3 zeros, $p = 0.4$ feels right. The MLE simply formalises this intuition. *Always think this way first.*

3.3 Gaussian Likelihood

- The density (pdf) of a single observation $X_i \sim N_p(\mu, \Sigma)$ is:

$$\varphi_{\mu, \Sigma}(x) = \frac{1}{\sqrt{(2\pi)^p |\Sigma|}} \exp \left[-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right]$$

where $|\Sigma| = \det(\Sigma)$ is the determinant of Σ .

- Since observations are iid, the joint **likelihood** factors as a product:

$$L(\mu, \Sigma \mid \text{data}) = \prod_{i=1}^n \varphi_{\mu, \Sigma}(X_i) = \left[\prod_{i=1}^n \frac{1}{\sqrt{(2\pi)^p |\Sigma|}} \right] \exp \left[-\frac{1}{2} \sum_{i=1}^n (X_i - \mu)^\top \Sigma^{-1} (X_i - \mu) \right]$$

- The **log-likelihood** is:

$$\ell(\mu, \Sigma \mid \text{data}) = -\frac{n}{2} \log[(2\pi)^p |\Sigma|] - \frac{1}{2} \sum_{i=1}^n (X_i - \mu)^\top \Sigma^{-1} (X_i - \mu)$$

- The **MLE optimisation problem** is:

$$(\hat{\mu}, \hat{\Sigma}) = \arg \max_{\substack{\mu \in \mathbb{R}^p \\ \Sigma \succ 0, p \times p}} \ell(\mu, \Sigma \mid \text{data})$$

where the constraint $\Sigma \succ 0$ means Σ must be positive definite (necessary for a valid covariance matrix — all eigenvalues strictly positive, hence Σ invertible).

3.4 MLE Results — Theorem

- Theorem:** Given $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N_p(\mu, \Sigma)$, the MLEs are:

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^\top = \frac{n-1}{n} S$$

- Bias analysis:**

- $\hat{\mu}$ is **unbiased**: $\mathbb{E}[\hat{\mu}] = \mu$. ✓
- $\hat{\Sigma}$ is **biased**: $\mathbb{E}[\hat{\Sigma}] = \frac{n-1}{n} \Sigma \neq \Sigma$. ✗
- S (dividing by $n-1$) is **unbiased**: $\mathbb{E}[S] = \Sigma$. ✓

- Key insight:** The MLE of Σ divides by n , not $n-1$. It systematically *underestimates* the true covariance. The difference is negligible for large n , but important in small samples.

- Why is $\hat{\Sigma}$ biased?** Because $\bar{X}$ is computed from the same data — this “uses up” one degree of freedom. The deviations $(X_i - \bar{X})$ are constrained to sum to zero, making them systematically smaller in magnitude than the true deviations $(X_i - \mu)$ would be.

- **Summary table:**

Estimator	Formula	Denominator	Bias
$\hat{\mu}$ (MLE)	$\frac{1}{n} \sum X_i$	n	Unbiased
$\hat{\Sigma}$ (MLE)	$\frac{1}{n} \sum (X_i - \bar{X})(X_i - \bar{X})^\top$	n	Biased
S (sample cov.)	$\frac{1}{n-1} \sum (X_i - \bar{X})(X_i - \bar{X})^\top$	$n-1$	Unbiased
Key link: $\hat{\Sigma} = \frac{n-1}{n} S$ and $\mathbb{E}[\hat{\Sigma}] = \frac{n-1}{n} \Sigma$			

3.5 Invariance Property of MLEs

- **Setup:** Let $\theta \in \mathbb{R}^k$ be any parameter (for the Gaussian: $\theta = (\mu, \Sigma)$), let $\hat{\theta}$ be its MLE, and let $h : \mathbb{R}^k \rightarrow \mathbb{R}^j$ be any function.

- **Invariance Property:**

$$\widehat{h(\theta)} = h(\hat{\theta})$$

The MLE of *any function* of θ is obtained by applying that function to the MLE of θ . You never need to re-derive.

- **Why does this matter?** Suppose you want the MLE of:
 - The correlation matrix (a function of Σ) $\Rightarrow$ compute from $\hat{\Sigma}$.
 - The determinant $|\Sigma| \Rightarrow$ result is $|\hat{\Sigma}|$.
 - The eigenvalues $\lambda_i(\Sigma) \Rightarrow$ result is $\lambda_i(\hat{\Sigma})$, the eigenvalues of $\hat{\Sigma}$.
 - Any other transformation $\Rightarrow$ just apply it to $\hat{\Sigma}$.
- **Invariance in practice — exam rule:** “Find the MLE of [some function of θ]” $\Rightarrow$ first find $\hat{\theta}$, then apply the function. Done.
- **Intuition:** The likelihood surface is the same regardless of parameterisation. The peak in θ -space corresponds to the peak in $h(\theta)$ -space (when h is bijective), so the MLE transforms cleanly.
- **Eigendecomposition example:** Any symmetric positive definite matrix Σ admits the **spectral decomposition**:

$$\Sigma = \sum_{i=1}^p \lambda_i \xi_i \xi_i^\top$$

where:

- $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p > 0$: the **eigenvalues** (real, positive scalars).
- $\xi_1, \dots, \xi_p \in \mathbb{R}^p$: the corresponding **eigenvectors**, orthonormal: $\xi_i^\top \xi_j = \delta_{ij}$.
- Each term $\lambda_i \xi_i \xi_i^\top$ is rank-1 — it captures variance in direction ξ_i .
- By the invariance property, the MLE of (λ_i, ξ_i) is $(\hat{\lambda}_i, \hat{\xi}_i)$ — the eigenvalues and eigenvectors of $\hat{\Sigma}$:

$$(\widehat{\lambda_i, \xi_i}) = (\hat{\lambda}_i, \hat{\xi}_i) \implies \hat{\Sigma} = \sum_{i=1}^p \hat{\lambda}_i \hat{\xi}_i \hat{\xi}_i^\top$$

- **Geometric intuition:** Think of Σ as a *variance ellipsoid*. The eigenvectors ξ_i are the **axes** of the ellipsoid; the eigenvalues λ_i are the **lengths** (variance in each direction). The largest eigenvalue λ_1 points in the direction of maximum variance — this is exactly the first principal component (PCA). The MLE simply estimates this ellipsoid from data.

4 Distribution of the Sample Mean

- Once we have an estimator, we need to know its **distribution**: what does $\bar{X}$ look like from sample to sample?
- **Theorem (Exact):** If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N_p(\mu, \Sigma)$, then for any $n \geq 1$:

$$\bar{X} \sim N_p\left(\mu, \frac{1}{n}\Sigma\right)$$

This is **exact** — not an approximation. It holds because Gaussians are closed under sums and scalar multiplication.

- **Intuition:**
 - The mean of Gaussian random vectors is again Gaussian (closure property).
 - Expected value stays μ : confirms $\bar{X}$ is unbiased.
 - Covariance shrinks by factor $1/n$: more data $\Rightarrow$ tighter concentration around μ .
 - Completely analogous to the univariate result $\bar{X} \sim N(\mu, \sigma^2/n)$.
- **Consequence — quadratic form and confidence region:** From the distribution of $\bar{X}$, it follows that:

$$n(\bar{X} - \mu)^\top \Sigma^{-1}(\bar{X} - \mu) \sim \chi_p^2$$

This is the multivariate analogue of the z -test. It defines an **ellipsoidal confidence region** for μ :

$$\mathcal{E} = \{\mu \in \mathbb{R}^p : n(\bar{X} - \mu)^\top \Sigma^{-1}(\bar{X} - \mu) \leq \chi_{p,\alpha}^2\}$$

The shape is an ellipse because Σ is not necessarily spherical — directions of high variance produce a wider ellipsoid; directions of low variance produce a narrower one.

5 The Wishart Distribution

5.1 Motivation — From χ^2 to Wishart

- We now know the distribution of $\bar{X}$. We need the distribution of the sample covariance matrix S .

- **Scalar analogy (recall from 1D statistics):** For $X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

This tells us how the sample variance fluctuates around σ^2 .

- In the multivariate world, $S^2 \in \mathbb{R}$ becomes $S \in \mathbb{R}^{p \times p}$. We need a **distribution over matrices**. That is the Wishart distribution.
- **Slogan:** *Wishart is the matrix χ^2 .*
- **Why do we need this?**
 - To construct confidence regions for Σ .
 - To build hypothesis tests comparing covariance matrices.
 - To derive Hotelling's T^2 test (the multivariate t -test).
 - To understand how much the estimated ellipsoid $\hat{\Sigma}$ wobbles from sample to sample.

5.2 Definition

- **Definition:** A $p \times p$ random symmetric positive definite matrix W has a **Wishart distribution** $W \sim W_p(m, \Sigma)$ with m degrees of freedom and scale matrix Σ if:

$$W = \sum_{j=1}^m Z_j Z_j^\top, \quad \text{where } Z_1, \dots, Z_m \stackrel{\text{iid}}{\sim} N_p(0, \Sigma)$$

- **Construction intuition:**
 - Each $Z_j Z_j^\top$ is a $p \times p$ outer product matrix of rank 1.
 - Summing m independent rank-1 matrices gives a random matrix.
 - When $m \geq p$, the sum is almost surely positive definite (full rank).
 - Just as $\chi_m^2 = \sum_{j=1}^m Z_j^2$ (sum of m squared scalars), the Wishart is a sum of m squared vectors in matrix form.

- **Special case $p = 1$:**

$$W_1(m, \sigma^2) = \sigma^2 \chi_m^2$$

When $p = 1$, the Wishart reduces exactly to a scaled χ^2 . The scalar chi-squared is a special case of the Wishart.

- **The scalar–matrix analogy:**

Scalar ($p = 1$)	Matrix ($p \geq 2$)
$Z_j \sim N(0, \sigma^2)$, a scalar	$Z_j \sim N_p(0, \Sigma)$, a vector
Z_j^2 is a number	$Z_j Z_j^\top$ is a $p \times p$ rank-1 matrix
$W = \sum_{j=1}^m Z_j^2$ (sum of scalars)	$W = \sum_{j=1}^m Z_j Z_j^\top$ (sum of outer products)
$W/\sigma^2 \sim \chi_m^2$	$W \sim W_p(m, \Sigma)$
Result: a single positive number	Result: a $p \times p$ positive definite matrix

5.3 The Key Theorem — Connection to the Sample Covariance

- **Theorem:** If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N_p(\mu, \Sigma)$, then:

$$(n-1)S = \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^\top \sim W_p(n-1, \Sigma)$$

- Moreover, $\bar{X}$ and S are **independent**.
- **Why $n-1$ degrees of freedom, not n ?**
 - The deviations satisfy the constraint $\sum_{i=1}^n (X_i - \bar{X}) = 0$ — always sum to zero.
 - This constraint removes one degree of freedom (exactly like in the scalar case).
 - We estimate $\bar{X}$ from the same data, consuming one degree of freedom.
 - Exactly analogous to $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$ in 1D.
- **Independence of $\bar{X}$ and S :**
 - One of the most important structural facts in multivariate statistics.
 - In 1D: independence of $\bar{X}$ and S^2 allows the t -test to work.
 - In p D: independence of $\bar{X}$ and S allows Hotelling's T^2 test.
 - **Intuition:** Knowing how the sample is spread around its centre (S) tells you nothing about where the centre is ($\bar{X}$) — at least for Gaussian data.

5.4 Main Properties of the Wishart Distribution

- **Property 1 — Expected value:**

$$W \sim W_p(m, \Sigma) \implies \mathbb{E}[W] = m\Sigma$$

This directly confirms our bias results:

- $\mathbb{E}[(n-1)S] = (n-1)\Sigma \Rightarrow \mathbb{E}[S] = \Sigma$ (unbiased). ✓
- $\mathbb{E}[\hat{\Sigma}] = \mathbb{E}\left[\frac{n-1}{n}S\right] = \frac{n-1}{n}\Sigma$ (biased). ✗
- **Property 2 — Additivity:** If $W_1 \sim W_p(m_1, \Sigma)$ and $W_2 \sim W_p(m_2, \Sigma)$ are independent, then:

$$W_1 + W_2 \sim W_p(m_1 + m_2, \Sigma)$$

Degrees of freedom add up — exactly like $\chi_{m_1}^2 + \chi_{m_2}^2 \sim \chi_{m_1+m_2}^2$.

- **Property 3 — Linear transformation:** For any matrix $A \in \mathbb{R}^{q \times p}$ with $\text{rank}(A) = q \leq p$:

$$AWA^\top \sim W_q(m, A\Sigma A^\top)$$

Wishart structure is preserved under linear maps. The dimension changes from p to q , and the scale matrix transforms as $\Sigma \mapsto A\Sigma A^\top$. Interpretation: if you project p -dimensional data onto a q -dimensional subspace, the projected covariance is still Wishart.

- **Property 4 — Scalar projection:** For any fixed non-zero vector $a \in \mathbb{R}^p$:

$$a^\top W a \sim (a^\top \Sigma a) \chi_m^2$$

Project the random matrix W onto any direction a and you recover a scaled χ^2 . This is the **bridge from matrix to scalar statistics**: it lets you apply 1D tests to specific linear combinations of the multivariate observations. (Special case of Property 3 with $A = a^\top$.)

- **Property 5 — Positive definiteness and invertibility:**

$$W \sim W_p(m, \Sigma) \text{ is almost surely positive definite} \iff m \geq p$$

- If $m < p$: W is almost surely singular ($\text{rank } m < p$) — it cannot be inverted.
- **Practical consequence:** Since $(n-1)S \sim W_p(n-1, \Sigma)$, if $n-1 < p$ (i.e. $n \leq p$), then S is singular. You cannot compute S^{-1} , so standard tests fail completely.
- **This is the curse of dimensionality:** If you have more variables p than observations n , classical multivariate methods break down.
- **Rule of thumb:** You need $n > p+1$ for S to be invertible. In practice, $n \gg p$ is required for reliable estimates.
- **Complete scalar–matrix analogy:**

Scalar ($p = 1$)	Matrix ($p \geq 2$)
$X_i \sim N(\mu, \sigma^2)$	$X_i \sim N_p(\mu, \Sigma)$
$\bar{X} \sim N(\mu, \sigma^2/n)$	$\bar{X} \sim N_p(\mu, \Sigma/n)$
$(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$	$(n-1)S \sim W_p(n-1, \Sigma)$
$\bar{X}$ and S^2 are independent	$\bar{X}$ and S are independent
$t\text{-test: } \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \sim t_{n-1}$	Hotelling's T^2 : $n(\bar{X} - \mu_0)^\top S^{-1}(\bar{X} - \mu_0)$

6 Law of Large Numbers (LLN)

6.1 Motivation

- The Wishart result is **exact but requires Gaussianity**. What if the data is not Gaussian?
- **Question:** Without assuming any particular distribution, can we still trust $\bar{X}$ as an estimate of μ ?
- **Answer (LLN):** Yes. $\bar{X}$ converges to μ regardless of the distribution, provided:
 - The data is iid.
 - The mean $\mu = \mathbb{E}[X_i]$ exists and is finite.
 - The covariance $\Sigma = \text{Cov}(X_i)$ is finite.
- **LLN is the foundation of why statistics works:** given enough data, your estimates are essentially correct — for any underlying distribution.

6.2 Setup

- Let $X_1, X_2, \dots \stackrel{\text{iid}}{\sim}$ any distribution with $\mathbb{E}[X_i] = \mu \in \mathbb{R}^p$ and $\text{Cov}(X_i) = \Sigma < \infty$.
- **Gaussianity is not required.** The results hold for Bernoulli, Poisson, uniform, t -distribution, and any other distribution with finite mean and variance.

6.3 Weak Law of Large Numbers (WLLN)

- **Statement:**

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu \quad \text{as } n \rightarrow \infty$$

Convergence in probability: for any $\varepsilon > 0$:

$$P(\|\bar{X}_n - \mu\| > \varepsilon) \xrightarrow{n \rightarrow \infty} 0$$

- **Interpretation:** The probability that $\bar{X}_n$ is far from μ (by any fixed distance ε) shrinks to zero as n grows.
- **Intuition:** Picture $\bar{X}_n$ as a dart landing near a bullseye μ . Early throws are scattered, but as n increases, the cluster centre converges to the bullseye. WLLN says the probability of a dart landing far away vanishes.

6.4 Strong Law of Large Numbers (SLLN)

- **Statement:**

$$\bar{X}_n \xrightarrow{a.s.} \mu \quad \text{as } n \rightarrow \infty$$

Almost sure (a.s.) convergence: for almost every possible infinite realisation of $X_1, X_2, \dots$:

$$\lim_{n \rightarrow \infty} \bar{X}_n = \mu$$

- **WLLN vs. SLLN:**
 - WLLN: “For each n , the probability of being far away is small.”
 - SLLN: “Almost certainly, you will eventually stay close forever.”
 - SLLN $\Rightarrow$ WLLN (stronger implication), but WLLN $\nRightarrow$ SLLN.
 - In practice, the distinction rarely matters for applications.

6.5 LLN for the Covariance Matrix

- The analogous result holds for the sample covariance:

$$S_n = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)(X_i - \bar{X}_n)^\top \xrightarrow{P} \Sigma \quad \text{as } n \rightarrow \infty$$

- Both $\bar{X}_n$ and S_n are **consistent estimators**: they converge in probability to the true parameters.
- **Consistency vs. unbiasedness:** $\hat{\Sigma}$ is biased for finite n , but consistent (it converges to Σ as $n \rightarrow \infty$). An estimator can be biased for every finite n yet still converge to the truth.

7 Central Limit Theorem (CLT)

7.1 Motivation

- The LLN tells us $\bar{X}_n \rightarrow \mu$. Two further questions remain:
 - **How fast** does it converge?
 - **What shape** does the error $\bar{X}_n - \mu$ have for finite (large) n ?
- The error $\bar{X}_n - \mu \rightarrow 0$ collapses to zero for large n , so examining it directly is uninformative. We **zoom in** by rescaling:

$$\sqrt{n}(\bar{X}_n - \mu)$$

This inflates the shrinking error at exactly the right rate. The factor $\sqrt{n}$ is correct because $\text{Var}(\bar{X}_n) = \Sigma/n$ shrinks at rate $1/n$, and multiplying by $\sqrt{n}$ exactly cancels this shrinkage.

7.2 Multivariate Central Limit Theorem

- **Theorem (Multivariate CLT):** Let $X_1, X_2, \dots \stackrel{\text{iid}}{\sim}$ any distribution with $\mathbb{E}[X_i] = \mu \in \mathbb{R}^p$ and $\text{Cov}(X_i) = \Sigma$ finite. Then:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N_p(0, \Sigma) \quad \text{as } n \rightarrow \infty$$

Equivalently, for large n :

$$\bar{X}_n \sim N_p\left(\mu, \frac{\Sigma}{n}\right)$$

- **Critical insight:** Gaussianity of the data is **not required**. The data can follow any distribution (Bernoulli, Poisson, uniform, etc.) provided Σ is finite. The CLT guarantees that the sample mean is approximately Gaussian for large n , regardless of the underlying distribution.
- **Why does the Gaussian appear everywhere in statistics?** Not because everything is Gaussian, but because *averages of anything become Gaussian*. The Gaussian is the universal attractor of sums and averages.
- **Connection to the exact result:** When the data *is* Gaussian, $\bar{X}_n \sim N_p(\mu, \Sigma/n)$ exactly for all n . The CLT says this exact result holds *approximately* for any distribution when n is large.

7.3 Rate of Convergence

- From the CLT, the typical magnitude of the error is:

$$\|\bar{X}_n - \mu\| \sim \frac{C}{\sqrt{n}}, \quad C \text{ depends on } \Sigma$$

- This $1/\sqrt{n}$ rate is a **fundamental law of statistical estimation**:
 - To halve the error $\Rightarrow$ need **4 times** as much data.

- To cut the error by 10 $\Rightarrow$ need **100 times** as much data.
- To cut the error by 100 $\Rightarrow$ need **10,000 times** as much data.
- This slow rate is why large datasets matter enormously in high-dimensional settings.

7.4 Practical Applications — Confidence Regions

- **When Σ is known** (or Gaussian data, exact result):

$$n(\bar{X}_n - \mu)^\top \Sigma^{-1}(\bar{X}_n - \mu) \sim \chi_p^2$$

Approximate $(1 - \alpha)$ confidence ellipsoid for μ :

$$\mathcal{E} = \{\mu : n(\bar{X}_n - \mu)^\top \Sigma^{-1}(\bar{X}_n - \mu) \leq \chi_{p, 1-\alpha}^2\}$$

- **When Σ is unknown** (replace by S , valid for large n):

$$n(\bar{X}_n - \mu)^\top S^{-1}(\bar{X}_n - \mu) \sim \chi_p^2$$

- **Important note:** When Σ is unknown and n is *small* with Gaussian data, the exact result is Hotelling's T^2 , which accounts for the additional uncertainty in estimating Σ .
- **Shape of the confidence region:** It is an ellipsoid because Σ encodes correlations between coordinates. Directions of high variance $\Rightarrow$ wider ellipsoid. Directions of low variance $\Rightarrow$ narrower ellipsoid.

7.5 Exact vs. Asymptotic — Full Comparison

	Gaussian data (any n)	Any distribution (large n)
Distribution of $\bar{X}_n$	Exactly $N_p(\mu, \Sigma/n)$	Approximately $N_p(\mu, \Sigma/n)$
Distribution of $(n-1)S$	Exactly $W_p(n-1, \Sigma)$	No simple exact form
$\bar{X}_n$ and S	Independent (exact)	Asymptotically independent
Confidence region	Exact ellipsoid (Hotelling T^2)	Approximate ellipsoid (χ_p^2)
Source of result	Closure of Gaussian	CLT

- **Exam rule of thumb:**
 - “Normally distributed data” $\Rightarrow$ use exact Wishart and χ^2 results.
 - “Large sample” or “general distribution” $\Rightarrow$ use CLT approximation.
 - Exact results are always stronger (valid for all n , not just large n).
 - When both apply (Gaussian + large n): exact result is preferred but CLT gives the same answer asymptotically.

7.6 Key Intuitions to Remember

- **Dart analogy for $\bar{X}_n$:** Each sample gives you a dart landing near the bullseye μ . LLN: the cluster converges to the bullseye. CLT: the cluster is shaped like a Gaussian ellipse with radius shrinking as $1/\sqrt{n}$.
- **Wishart = uncertainty of a shape:** S is not a number — it is a whole ellipsoid. The Wishart distribution describes how this ellipsoid wobbles from sample to sample. More degrees of freedom $n - 1 \Rightarrow$ less wobble, more stable estimate.
- **The $n > p$ barrier:** If $n \leq p$, the sample covariance S is singular. You cannot invert it. Standard MLE and hypothesis tests fail. This is the core challenge of modern high-dimensional statistics.
- **Independence of $\bar{X}$ and S :** The single most important structural fact for hypothesis testing. Without it, neither the t -test nor Hotelling's T^2 would exist. For Gaussian data: always independent.
- **The Gaussian is universal:** It appears not because nature is Gaussian, but because the CLT forces averages toward it. This is the deepest reason the normal distribution is central to all of statistics.

Türkçe Açıklamalar / Turkish Explanations

1. Büyük Resim ve Motivasyon

- $X_1, \dots, X_n$ verilerini gözlemliyoruz ve bu verilerin **bilinmeyen** parametreler μ ve Σ ile bir Gaussian dağılımdan geldiğine inanıyoruz.
- **Temel soru:** μ ve Σ 'yı sonlu bir örneklemden nasıl tahmin ederiz? Tahminlerimiz ne kadar iyi?
- Ders dört adımda ilerler:
 1. **Kurulum:** Modeli ve sezgisel “açık” tahmin edicileri tanımla.
 2. **MLE:** Maksimum olabilirlik ilkesinden tahmin edicileri türet.
 3. **Kesin dağılımlar:** Gaussianlık varsayımı altında $\bar{X}$ ve S 'nin dağılımını bul (Wishart gerektirir).
 4. **Asimptotik:** $n \rightarrow \infty$ iken ne olduğunu açıkla (BBS, MST) — Gaussianlık gerekmez.
- **Temel ayırım:**
 - **Kesin sonuçlar** (Wishart, χ^2): herhangi bir n için geçerli, Gaussian veri gerektirir.
 - **Asimptotik sonuçlar** (BBS, MST): yalnızca sonlu ortalama ve kovaryans gerektirir, herhangi bir dağılım için geçerlidir.
 - *Sınav kuralı:* “Gaussian dağılımlı veri” $\Rightarrow$ kesin sonuçlar. “Büyük örneklem” $\Rightarrow$ MST/BBS.

2. Kurulum — Çok Değişkenli Gaussian Modeli

- $\underline{X} \sim N_p(\mu, \Sigma)$: ortalama $\mu \in \mathbb{R}^p$ ve kovaryans Σ (simetrik, pozitif tanımlı) olan p -boyutlu rastgele vektör.
- n adet iid gözlem: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N_p(\mu, \Sigma)$
- **Hedef:** Bilinmeyen μ ve Σ 'yı yalnızca veriden tahmin et.
- **Açık tahmin ediciler:**
 - Örneklem ortalaması (tarafsız): $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, $\mathbb{E}[\bar{X}] = \mu$
 - Örneklem kovaryansı (tarafsız): $S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^\top$, $\mathbb{E}[S] = \Sigma$
 - $n - 1$ paydası **Bessel düzeltmesidir**: $\bar{X}$ aynı veriden hesaplandığından sapmaların sistematik küçüklüğünü telafi eder.
- **Sezgi:** 1D'deki $\bar{X}$ ve S^2 'nin doğrudan çok boyutlu karşılıklarıdır.

3. Maksimum Olabilirlik Tahmini (MLE)

- Prensipli, genel bir yöntem istiyoruz. **MLE ilkesi:** Gözlemlenen veriyi en olası kılan parametre değerini seç.

$$\hat{\theta} = \arg \max_{\theta} L(\theta \mid \text{veri})$$

- Uygulamada **log-olabilirlik** $\ell = \log L$ maksimize edilir: çarpımlar toplamlar olur, türev almak kolaylaşır, argmax değişmez.
- **Bernoulli ısınma örneği:** $X_1 = 0, X_2 = 0, X_3 = 1, X_4 = 0, X_5 = 1$ için:

$$L(p) = p^2(1-p)^3, \quad \ell(p) = 2 \log p + 3 \log(1-p), \quad \hat{p} = \frac{2}{5} = \bar{X}$$

Sezgi: Formülden önce şunu sor — “hangi p değeri bu veriyi en olası kılar?” 2 başarı, 3 başarısızlık varsa $p = 0.4$ sezgisel olarak doğru.

- **Gaussian log-olabilirliği:**

$$\ell(\mu, \Sigma) = -\frac{n}{2} \log[(2\pi)^p |\Sigma|] - \frac{1}{2} \sum_{i=1}^n (X_i - \mu)^\top \Sigma^{-1} (X_i - \mu)$$

- **MLE Teoremi:**

$$\hat{\mu} = \bar{X} \quad (\text{tarafsız: } \mathbb{E}[\hat{\mu}] = \mu)$$

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^\top = \frac{n-1}{n} S \quad (\text{tarafı: } \mathbb{E}[\hat{\Sigma}] = \frac{n-1}{n} \Sigma)$$

- **Neden $\hat{\Sigma}$ tarafı?** $\bar{X}$ aynı veriden hesaplanır ve bir serbestlik derecesi tüketir. Sapmaların toplamı sıfır olmak zorundadır ($\sum (X_i - \bar{X}) = 0$), bu da onları sistematik olarak küçültür.
- **Tutarlılık vs. tarafsızlık:** $\hat{\Sigma}$ tarafıdır ama tutarlıdır — yani $n \rightarrow \infty$ için Σ 'ya yakınsar.
- **Değişmezlik Özelliği:** $\widehat{h(\theta)} = h(\hat{\theta})$
 - θ 'nın MLE'si $\hat{\theta}$ ise, her h fonksiyonu için $h(\theta)$ 'nın MLE'si $h(\hat{\theta})$ 'dır.
 - Yeniden türetme gerekmez. Korelasyon matrisi, determinant, özdeğerler — hepsi $\hat{\Sigma}$ 'dan elde edilir.
 - **Sınav sezgisi:** “Bir şeyin fonksiyonunun MLE'si” sorusunda ilk düşünce her zaman değişmezlik olmalı.

- **Eigendecomposition örneği:**

$$\Sigma = \sum_{i=1}^p \lambda_i \xi_i \xi_i^\top \quad \text{değişmezlik} \quad \hat{\Sigma} = \sum_{i=1}^p \hat{\lambda}_i \hat{\xi}_i \hat{\xi}_i^\top$$

$\hat{\lambda}_i$ ve $\hat{\xi}_i$, $\hat{\Sigma}$ 'nin özdeğer ve özvektörleridir.

- **Geometrik sezgi:** Σ 'yı bir *varyans elipsoidi* olarak düşün. Özvektörler elipsoidin eksenleri, özdeğerler o eksenlerdeki uzunluklardır (varyans miktarı). En büyük λ_1 , verinin en çok yayıldığı yön — PCA'nın ta kendisi. MLE bu elipsoidi veriden tahmin eder.

4. Örneklem Ortalamasının Dağılımı

- Bir tahmin edici bulduktan sonra **dağılımını** bilmek gerekir: $\bar{X}$ örneklemden örnekleme nasıl değişiyor?
- **Teorem (Kesin):** $X_i \stackrel{\text{iid}}{\sim} N_p(\mu, \Sigma)$ ise, herhangi bir $n \geq 1$ için:

$$\bar{X} \sim N_p\left(\mu, \frac{1}{n}\Sigma\right)$$

Bu bir yaklaşım değil — **kesin sonuçtur**. Gaussianlar toplamalar altında kapalıdır.

- **Sezgi:**
 - Gaussian vektörlerin ortalaması yine Gaussian (kapanma özelliği).
 - Beklenen değer μ kalır: tarafsızlığı doğrular.
 - Kovaryans $1/n$ ile küçülür: daha fazla veri $\Rightarrow \mu$ etrafında daha sıkı yoğunlaşma.
 - 1D sonuç $\bar{X} \sim N(\mu, \sigma^2/n)$ 'nin tam karşılığı.

- **Güven bölgesi:**

$$n(\bar{X} - \mu)^\top \Sigma^{-1}(\bar{X} - \mu) \sim \chi_p^2$$

μ için eliptik güven bölgesi tanımlar. Σ küresel değilse bölge de küre değildir.

5. Wishart Dağılımı

- $\bar{X}$ 'in dağılımını biliyoruz. Şimdi S 'nin dağılımına ihtiyacımız var.
- **1D analoji:** $X_i \sim N(\mu, \sigma^2)$ için $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$. Çok boyutlu dünyada $S \in \mathbb{R}^{p \times p}$ için matrisler üzerinde bir dağılıma ihtiyaç duyulur.
- **Slogan:** *Wishart, matrisler için χ^2 'dir.*
- **Neden gerekli?** Σ için güven bölgeleri, kovaryans testi, Hotelling T^2 testi için.
- **Tanım:**

$$W = \sum_{j=1}^m Z_j Z_j^\top, \quad Z_j \stackrel{\text{iid}}{\sim} N_p(0, \Sigma) \quad \Rightarrow \quad W \sim W_p(m, \Sigma)$$

- Her $Z_j Z_j^\top$: rank-1'lik $p \times p$ dış çarpım matrisi.
- m adet bağımsız rank-1 matrisin toplamı.
- $m \geq p$ ise neredeyse kesinlikle pozitif tanımlı.
- $p = 1$ için: $W_1(m, \sigma^2) = \sigma^2 \chi_m^2$ — tam olarak ölçeklenmiş χ^2 .

- **Temel Teorem:**

$$(n-1)S \sim W_p(n-1, \Sigma), \quad \bar{X} \text{ ve } S \text{ bağımsızdır.}$$

- **Neden $n-1$?** $\sum(X_i - \bar{X}) = 0$ kısıtı bir serbestlik derecesi tüketir. 1D durumundaki χ_{n-1}^2 ile birebir aynı neden.
- **$\bar{X}$ ve S 'nin bağımsızlığı:** Tüm hipotez testlerini mümkün kılan yapısal gerçek. 1D'de t -testi için, p D'de Hotelling T^2 testi için zorunludur. **Ezberle.**

- **Ana özellikler:**

- **Beklenen değer:** $\mathbb{E}[W] = m\Sigma \Rightarrow \mathbb{E}[S] = \Sigma$ (tarafsız), $\mathbb{E}[\hat{\Sigma}] = \frac{n-1}{n}\Sigma$ (tarafı)
- **Toplanabilirlik:** $W_1 + W_2 \sim W_p(m_1 + m_2, \Sigma)$ (tıpkı χ^2 gibi)
- **Doğrusal dönüşüm:** $AWA^\top \sim W_q(m, A\Sigma A^\top)$ (Wishart yapısı korunur)
- **Skaler projeksiyon:** $a^\top W a \sim (a^\top \Sigma a)\chi_m^2$ (matristeki istatistikten skalere köprü)
- **Pozitif tanımlılık:** W tersinebilir $\iff m \geq p$

- **Kritik pratik sonuç:** $n \leq p$ ise S tekildir, tersi alnamaz, standart testler çöker. Yüksek boyutlu istatistiğin temel sorunudur. Pratik kural: $n \gg p$ gereklidir.
- **Sınav sezgisi — Wishart = şeklin belirsizliği:** S bir sayı değil, tüm bir elipsoiddir. Wishart bu elipsoidin örneklemiden örnekleme nasıl sallandığını tanımlar. $n - 1$ büyük $\Rightarrow$ daha az sallantı, daha güvenilir tahmin.

6. Büyük Sayılar Yasası (BBS)

- Wishart kesin ama Gaussianlık gerektirir. Dağılım Gaussian değilse ne olur?
- **BBS cevabı:** $\bar{X}_n$, herhangi bir dağılım için μ 'ya yakınsar — yeter ki iid olsun ve ortalama/kovaryans sonlu olsun. Gaussianlık gerekmez.
- **BBS, istatistiğin neden işe yaradığının temelidir.**
- **Zayıf BBS (WLLN):**

$$\bar{X}_n \xrightarrow{P} \mu \iff P(\|\bar{X}_n - \mu\| > \varepsilon) \rightarrow 0 \quad \forall \varepsilon > 0$$

$\bar{X}_n$ 'in μ 'dan uzakta olma olasılığı her sabit ε için sıfıra gider.

- **Güçlü BBS (SLLN):**

$$\bar{X}_n \xrightarrow{a.s.} \mu \iff P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1$$

Neredeyse her sonsuz veri dizisi için $\bar{X}_n \rightarrow \mu$. SLLN $\Rightarrow$ WLLN (daha güçlü).

- **Kovaryans için BBS:**

$$S_n \xrightarrow{P} \Sigma \quad (n \rightarrow \infty)$$

- $\bar{X}_n$ ve S_n **tutarlı tahmin edicilerdir:** gerçek parametreye olasılıkla yakınsar.
- **Tutarlılık vs. tarafsızlık:** $\hat{\Sigma}$ sonlu n için tarafı ama tutarlıdır.

7. Merkezi Limit Teoremi (MST)

- BBS $\bar{X}_n \rightarrow \mu$ diyor. Peki **ne kadar hızlı?** Ve $\bar{X}_n - \mu$ 'nun **şekli** nedir?
- $\bar{X}_n - \mu \rightarrow 0$ büyük n için tek başına ilginç değil. $\sqrt{n}$ ile ölçeklendiriyoruz:

$$\sqrt{n}(\bar{X}_n - \mu)$$

Neden $\sqrt{n}$? $\text{Var}(\bar{X}_n) = \Sigma/n$ hızıyla küçülüyor; $\sqrt{n}$ ile çarpmak tam olarak dengeler.

- **Çok Değişkenli MST:** $\mathbb{E}[X_i] = \mu$, $\text{Cov}(X_i) = \Sigma$ sonlu olmak üzere (herhangi bir dağılım):

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N_p(0, \Sigma) \quad (n \rightarrow \infty)$$

Eşdeğer: büyük n için $\bar{X}_n \sim N_p(\mu, \Sigma/n)$.

- **Kritik nokta:** Verinin Gaussian olması **gerekmez**. Bernoulli, Poisson, düzgün — fark etmez. Örneklem ortalaması büyük n için her zaman yaklaşık Gaussian olur.
- **Gaussian neden her yerde görünür?** Her şey Gaussian olduğu için değil; *herhangi bir şeyin ortalamaları Gaussian'a dönüştüğü* için. Gaussian, toplamların evrensel çekicisidir.
- **Kesin sonuçla bağlantı:** Veri Gaussian ise $\bar{X}_n \sim N_p(\mu, \Sigma/n)$ tam olarak tüm n için geçerli. MST bu kesin sonucun *yaklaşık* olarak herhangi bir dağılım için büyük n 'de geçerli olduğunu söyler.
- **Yakınsama hızı — temel tahmin yasası:**

$$\|\bar{X}_n - \mu\| \sim \frac{1}{\sqrt{n}}$$

- Hatayı yarıya indirmek için **4 kat** daha fazla veri gerekir.
- Hatayı 10'a bölmek için **100 kat** daha fazla veri gerekir.
- Hatayı 100'e bölmek için **10.000 kat** daha fazla veri gerekir.

- **Güven bölgesi uygulamaları:**

- Σ biliniyorsa: $n(\bar{X}_n - \mu)^\top \Sigma^{-1}(\bar{X}_n - \mu) \sim \chi_p^2$
- Σ bilinmiyorsa, S ile değiştir: $n(\bar{X}_n - \mu)^\top S^{-1}(\bar{X}_n - \mu) \sim \chi_p^2$
- Gaussian + küçük n : Hotelling T^2 (kesin sonuç, Σ bilinmemesi hesaba katılır).

- **Sınav kuralları:**

- “Gaussian dağılımlı veri” $\Rightarrow$ kesin Wishart ve χ^2 sonuçları kullan.
- “Büyük örneklem” veya “genel dağılım” $\Rightarrow$ MST kullan.
- Kesin sonuçlar her zaman daha güçlüdür (tüm n için geçerli).
- Her ikisi geçerliyse: kesin sonuç tercih edilir.

- **Profesörün vurguladığı sezgiler:**

- **Dart sezgisi:** Her örneklem $\bar{X}_n$ 'i bir dart olarak düşün. BBS: kümenin merkezi bullseye (μ) yakınsar. MST: küme Gaussian elips şeklindedir, yarıçapı $1/\sqrt{n}$ ile küçülür.
- **$n > p$ bariyeri:** $n \leq p$ ise S tekildir, tersi alınamaz, testler çöker. Yüksek boyutlu istatistiğin temel problemi budur.
- **$\bar{X}$ ve S 'nin bağımsızlığı:** En önemli yapısal gerçek. Gaussian veri için her zaman bağımsızdır.
- **Wishart = şeklin belirsizliği:** S bir sayı değil, bir elipsoiddir. Wishart bu elipsoidin sallantısını tanımlar.
- **$1/\sqrt{n}$ hızı:** İstatistiksel tahminin temel yasasıdır. Hatayı yarıya indirmek için 4 kat veri gerekir.